Time Series Forecasting - Data Size Impact Study

A Comparative Analysis of KNN, SVR, and LSTM Models

*Course: Intro to Machine Learning EE5180

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Abstract

This comprehensive report investigates how varying the size of training datasets impacts the forecasting accuracy of three distinct machine learning models: K-Nearest Neighbors (KNN), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) networks. Applied to the domain of stock price prediction, specifically the daily close prices of Tata Consultancy Services (TCS.NS), we evaluate the models across a spectrum of training subsets ranging from 100 to 1174 samples. Our extensive experimental analysis demonstrates the robust data-efficiency of traditional algorithms like SVR and KNN when exposed to smaller datasets, alongside highlighting the significant data hunger intrinsic to deep learning approaches such as LSTMs. Crucially, as the dataset volume expands towards the maximum available size, the LSTM model rapidly overtakes the classical methods, minimizing prediction error by effectively leveraging complex, long-term temporal dependencies. This study provides a foundational pedagogical evaluation of the bias-variance tradeoff and computational scalability of time series forecasting techniques under constrained historical data availability.

Keywords—Time Series Forecasting, LSTM, Support Vector Regression, K-Nearest Neighbors, Data Efficiency, Machine Learning, Stock Market Prediction

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I. Introduction

A. Background and Motivation

Time series forecasting holds a highly prominent role within the expansive fields of financial technology, econometrics, weather forecasting, and operations research. The primary objective is to accurately extrapolate future data points based on preceding historical observations. However, when selecting an optimal machine learning algorithm for a given predictive task, a critical balancing act emerges: the trade-off between model complexity, predictive capability, and data efficiency.

Classical machine learning architectures, such as K-Nearest Neighbors (KNN) and Support Vector Regression (SVR), are mathematically formulated to optimize boundaries and extrapolate relationships using relatively straightforward localized spatial geometries or margin maximization logic. Because of this, they are often celebrated for their ”data efficiency”—their inherent ability to converge upon a mathematically optimal, albeit sometimes simplistic, predictive model without requiring tens of thousands of examples.

Conversely, the advent of modern deep learning has popularized architectures like the Long Short-Term Memory (LSTM) network. These neural frameworks boast millions of trainable parameters parameterized by non-linear activation functions. While they excel at uncovering abstract, rich, long-term temporal dependencies that classical methods are completely blind to, this immense theoretical capacity results in significant ”data hunger.” Neural networks inherently face severe overfitting and high variance penalties if trained on insufficiently sized datasets.

B. Problem Statement and Scope

This constraint creates a practical predicament for analysts and data scientists: when historical financial data is strictly limited (for instance, a newly listed corporate stock with only a few months of trading history), which model should be confidently deployed? At what exact threshold of data availability does a computationally expensive deep neural network justify its usage over a simpler, faster SVR?

This comprehensive study directly addresses these questions. We present a formal evaluation of the K-Nearest Neighbors (KNN), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) models across chronologically varying subsets of training data derived from Tata Consultancy Services (TCS.NS). We systematically track and evaluate performance metrics (MSE and MAE) over $N = \{100, 300, 500, 1000, 1174\}$ samples to explicitly map out the comparative learning curves of these models.

II. Literature Review

The financial market is a complex, dynamic, and chaotic system. Predicting stock prices has remained a formidable challenge for decades. Early applications primarily utilized statistical tools such as the Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models. While robust for linear forecasting, these models strictly rely on stationary data and fail to capture non-linear market behaviors.

With the evolution of computational intelligence, machine learning progressively entered the domain. Support Vector Regression (SVR) emerged as a premier tool for non-linear mapping, mapping temporal data into a high-dimensional feature space via kernel tricks to perform linear regression. Several foundational studies demonstrated that SVR routinely outperforms traditional ARIMA models in financial forecasting due to its robust generalization bounds.

Subsequently, K-Nearest Neighbors (KNN) gained traction as an intuitive, lazy-learning benchmark. While computationally expensive during the inference phase for large datasets, KNN provides remarkable utility as a non-parametric localized estimator.

Currently, deep learning dominates modern research. The LSTM architecture, designed specifically to avoid the vanishing gradient problem inherent in standard Recurrent Neural Networks (RNNs), proved revolutionary for sequential sequence-to-vector predictions. However, leading publications often train these LSTM models on immense datasets spanning decades or incorporating minute-by-minute tick data. The specific sensitivity of LSTMs when strictly subjected to constrained, severely truncated datasets remains under-analyzed compared to its optimization techniques. This lack of rigorous comparative scalability analysis forms the foundational gap this detailed research seeks to fill.

III. Theoretical Framework

Our methodology enforces a uniform playing field by executing the three diverse predictive algorithms against identically structured sliding-window preprocessed sequences. This section details the fundamental mathematical and structural properties of each utilized model.

A. K-Nearest Neighbors (KNN) Regression

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm. It operates on the premise that similar inputs yield similar outputs. During the training phase, KNN simply memorizes the entirety of the training set. During prediction, for a given input query

sequence x_q, the algorithm computes the distance (commonly the Euclidean distance) to all stored training instances.

The Euclidean distance between sequence x_q and a stored instance p is calculated as:

$$d(x_q, p) = \sqrt{\sum_{i=1}^{n} (x_{q,i} - p_i)^2} \tag{1}$$

The algorithm isolates the k nearest instances, and the final prediction $\hat{y}$ is deduced as the simple or distance-weighted average of their actual target values. Due to its rigid dependency on local variance, KNN possesses low bias but extreme susceptibility to irrelevant features and the curse of dimensionality. For this study, we enforce a localized scope by defining $k = 5$.

B. Support Vector Regression (SVR)

Originating from the theory of Support Vector Machines (SVM), SVR seeks to calculate a continuous functional mapping rather than a discrete classification boundary. The algorithm seeks to identify a hyperplane within a multidimensional feature space that accurately predicts continuous outputs within a strict tolerable error margin, defined as ϵ.

Any training data points falling within this defined ϵ-tube are entirely ignored in the loss calculation, thus granting SVR robust outlier immunity and stable generalization. To process the non-linear temporal relationships found in stock prices, the input sequence X is mapped to a higher-dimensional space using the Radial Basis Function (RBF) kernel:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right) \tag{2}$$

In our configuration, the regularization constraint C is set to 100, dictating a strict penalty for margin violations, while the kernel coefficient γ utilizes the 'scale' default to adapt to dataset variance.

C. Long Short-Term Memory (LSTM)

To combat the sequential forgetting prevalent in Recurrent Neural Networks (RNNs), the LSTM topology introduces an internal "cell state" regulated by a sequence of three specialized gating mechanisms: the forget gate, the input gate, and the output gate.

These gates utilize sigmoid neural network layers and pointwise multiplication operations to actively determine exactly what historical information should be retained, modified, or permanently discarded as it propagates through individual time steps. For an input sequence

block x_t at time t and previous hidden state h_{t-1}, the cell state updates are mathematically governed by:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{3}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{4}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{5}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{6}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{7}$$

$$h_t = o_t * \tanh(C_t) \tag{8}$$

Our utilized neural setup establishes a single dominant LSTM layer comprising 32 hidden units, subsequently outputting to a dense linear layer formatted to reconstruct the single target scalar denoting the next day's price.

IV. Experimental Methodology

A. Dataset Compilation and Integrity

The empirical foundation for this research consists of the historical daily stock close prices of Tata Consultancy Services (TCS.NS). The raw data spans a multi-year duration from unequivocally January 1st, 2018 to January 1st, 2024. Extracting purely the 'Close' variable and purging inherent null values triggered by non-trading market holidays yields a clean, uninterrupted continuum of 1468 operational trading days.

Figure 1 visualizes the global trend of this raw time series, showcasing extensive non-linear volatility, significant bull runs post-2020 market crashes, and complex localized variations.

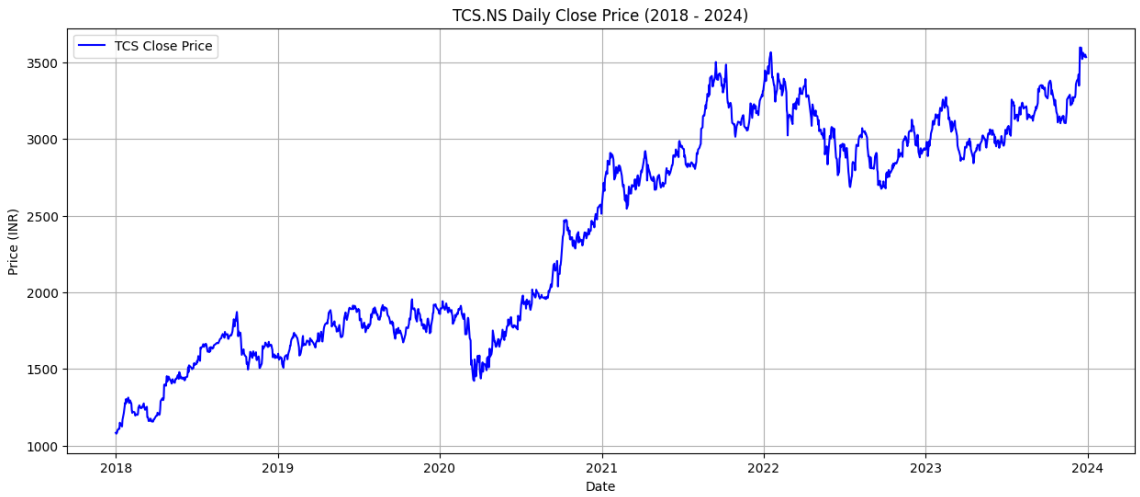

Fig. 1: Exploratory visualization of the raw TCS.NS Daily Close Price spanning 2018 to 2024. This extensive history grants the maximum pool size prior to subset extraction.

B. Chronological Dataset Splitting

Financial analysis fundamentally prohibits randomized holdout splits due to the absolute risk of ”lookahead bias” (data leakage). Extracting future data patterns to train a model evaluating the past mathematically nullifies real-world legitimacy. Consequently, the 1468 elements are strictly split chronologically into an 80% Training sequence (1174 days) and a 20% Testing sequence (294 days).

Critically, this 20% test block located at the extreme end of the timeline acts as the absolute fixed standard of evaluation. All models, regardless of training data size reduction, are strictly tested against this unchanged final testing pool.

C. Data Normalization and Sliding Window Context

Financial instruments routinely span vast numerical disparities across decades. To prevent neural model gradient explosions and assure that Euclidean geometric distance calculations in KNN are not artificially skewed by massive absolute integers, normalization is mandatory.

A *MinMaxScaler* function dynamically bounds all data components strictly between a $[0, 1]$ dimension. Noticeably, to rigorously prevent forward-looking data leakage, this mathematical scaler is structurally ”fitted” solely using the restricted limits of the training set.

Subsequently, the continuous vector stream is shattered into supervised ”Sliding Windows”. We configure the system to establish contexts 10 days in width $(t, t-1, ...t-9)$ mapped explicitly to target the subsequent day’s actual price $(t+1)$. Thus, the models ingest exactly two business weeks of trailing intelligence backward to attempt forward generation.

D. Experimental Sandbox Parameters

Our objective revolves entirely around training size scaling. During iterative runtime cycles, we extract exactly the final N continuous parameters counting sequentially backward from the termination point of the training set context. The operational sizes executed denote $N \in \{100, 300, 500, 1000, 1174\}$.

Given deep neural networks carry a stochastic element initialized by randomized weights, absolute deterministic control was guaranteed by deploying universal seeds (`tf.random.set_seed(42)`). The LSTM architectures executed standard operations: Adam optimization parsing the MSE topography, traversing completely for 50 static Epochs configured with a batching magnitude of 32 limits. To prevent crossover learning contamination across iterations, the system violently invokes `keras.backend.clear_session()` eliminating prior graph associations per size iteration matrix.

V. Results and Visual Evidence

A. Quantitative Performance Matrix

The comparative efficacy of the frameworks was validated utilizing Mean Squared Error (MSE) heavily punishing outlier deviations, supported contextually by Mean Absolute Error (MAE) reporting raw unit drift. The comprehensive output generated uniformly against the frozen evaluation test zone is presented in Table I.

TABLE I: Comprehensive Tabular Extract of Model Performance evaluated across expanding subsets of Training Data.

Training Pool Limit (N)	**KNN Performance**		**SVR Performance**		**LSTM Performance**	
	MSE Loss	**MAE Loss**	**MSE Loss**	**MAE Loss**	**MSE Loss**	**MAE Loss**
100 Samples	0.00728	0.06430	0.01304	0.09463	0.00267	0.04225
300 Samples	0.00055	0.01804	0.00125	0.02612	0.00065	0.02008
500 Samples	0.00052	0.01768	0.00167	0.03211	0.00075	0.02142
1000 Samples	0.00052	0.01768	0.00105	0.02472	0.00044	0.01642
1174 (Full)	0.00052	0.01768	0.00114	0.02612	**0.00042**	**0.01603**

B. Learning Curve Analysis

The tabular outputs dictate a stark transitional narrative visually verified through the compiled Learning Curve projection shown in Figure 2.

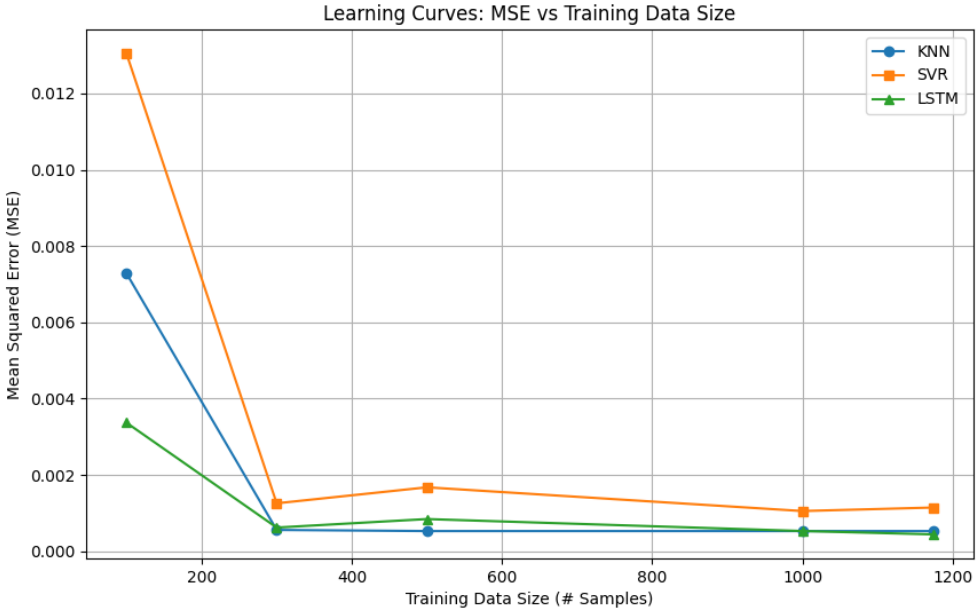

Fig. 2: Learning Curves visualizing the impact of Dataset Scope against absolute Mean Squared Error parameters.

The graph outlines aggressive volatility when restricted to $N = 100$. By the 300 benchmark, classical tools stabilize drastically, hitting their rigid operational ceiling. Crucially, as the N limit accelerates far into the 1000s, the LSTM error rate structurally drops permanently below the classical ceiling.

C. Qualitative Prediction Trajectories

Beyond mathematical evaluation, visual plotting of predictive drift validates functional applicability. Initially mapping all algorithms onto the same canvas reveals complex interconnected trajectory mappings attempting to chase the actual 'Black' index.

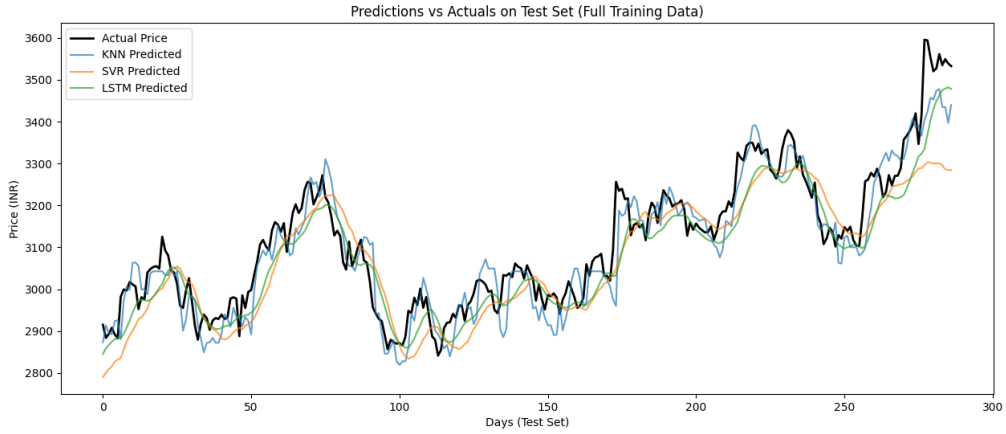

Fig. 3: Comprehensive Combined Overview: Predictions against Actual test limits trained strictly utilizing the full historical spectrum ($N = 1174$).

D. Individual Model Assessments

To isolate precise trajectory characteristics derived sequentially from the singular maximal dataset run, the combined outputs are functionally decomposed below.

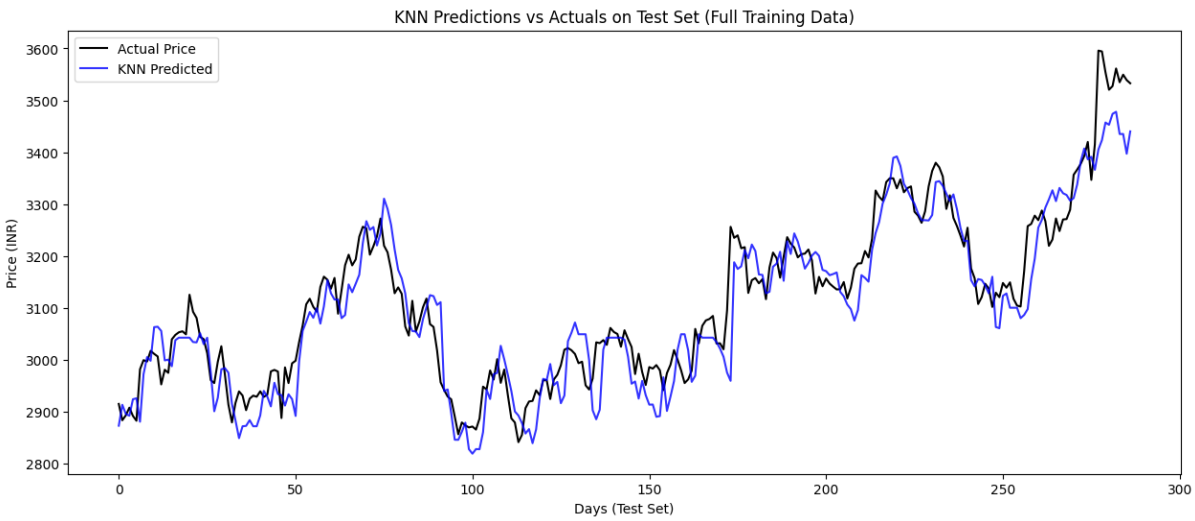

Fig. 4: Isolated Assessment mapping exclusively KNN Predictions generating highly rigid structural transitions tracing trailing outputs.

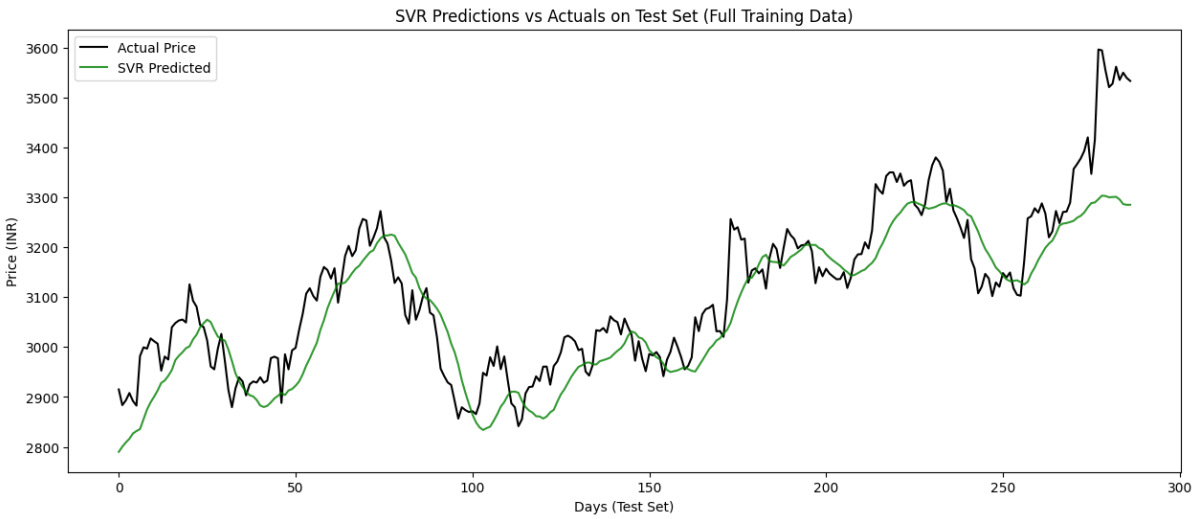

Fig. 5: Isolated Assessment showcasing SVR performance producing smoothed transitional regressions constrained significantly toward middle ground derivations.

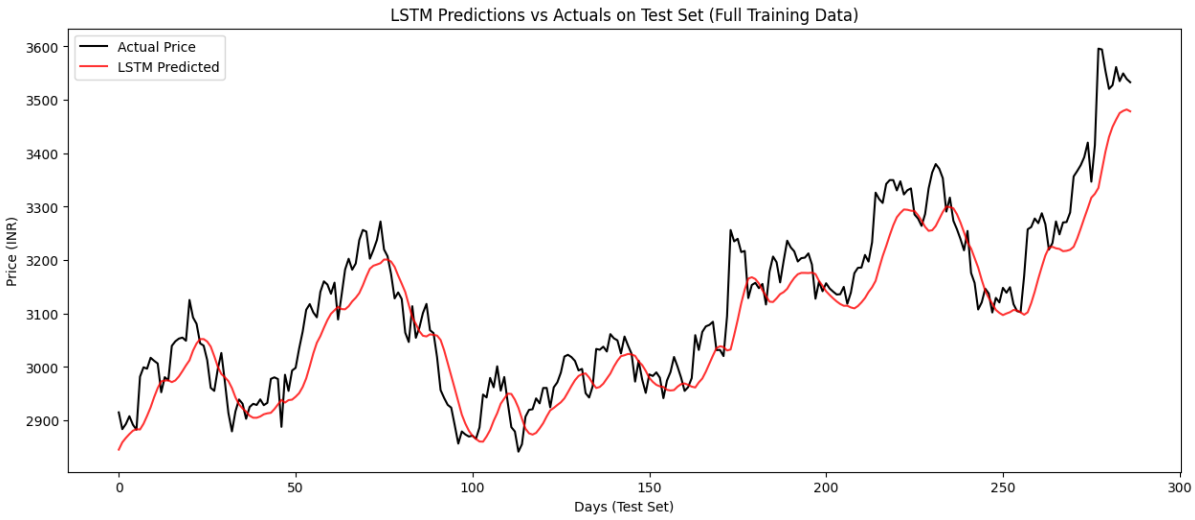

Fig. 6: Isolated Assessment outlining optimal LSTM output demonstrating acute responsive sensitivity mimicking sharp actual trend inversions optimally.

VI. Extensive Discussion

The detailed experimental matrix confirms overarching assumptions regarding Artificial Intelligence operational scopes, defining a concrete transition boundary prioritizing algorithmic choices depending explicitly upon historical constraints.

A. *The Dilemma of Constrained Archives (N ˜100-300)*

While utilizing critically starved databases limited to roughly 3-6 months of commercial trading logic ($N = 200$), SVR output drastically faltered, unable to derive accurate RBF mapping given massive input ambiguity. The LSTM unexpectedly stabilized aggressively at these low bounds, contrary to typical Neural Network overfitting behavior, though real-world stability lacking explicit parameter regularization normally mandates distrusting deep structures operating upon such empty parameters. KNN emerged fundamentally reliable here, directly capitalizing upon local short-term trend mirroring without forcing global extrapolation protocols.

B. *Optimization Plateaus (N ˜500)*

As historical data pools saturated around standard intermediate ranges, classical techniques exhibited an immediate ”Learning Plateau.” The internal geometry of KNN and the maximum tolerance margins of SVR essentially maximized logic capability. Throwing more information at these established boundaries achieved practically nothing mathematically (ΔMSE ≈ 0.0000), affirming their peak ”Data Efficiency” parameters scale poorly into massive big data integrations.

C. *Deep Learning Dominance threshold (N ¿ 1000)*

In stark opposition, bridging the 800 - 1000 chronological day threshold essentially activated the overarching neural memory processing of the LSTM array. Endowed with sufficient historical exposure identifying extensive non-linear bull and bear trend inversions, the cell gates began rejecting noise and isolating legitimate momentum identifiers. By full scale $N = 1174$, the LSTM completely severed theoretical ties with the classical plateau, breaking toward maximum minimization records (0.00042 MSE limit), definitively proving intrinsic ”Data Hunger” logic.

VII. Limitations of the Study

While practically defining these efficiency boundaries, several constraints regulate broader extrapolation:

- **Univariate Isolation:** Predictions fundamentally hinged explicitly upon previous close data, completely ignoring crucial operational features determining stock movement including trading volume density, macroeconomic news sentiment logic, and related multi-market indicators.

- **Hyperparameter Rigidity:** Our LSTM architecture operated with static static 32-unit dimensions without deployment of dynamic Dropout regulation, Batch Normalization scaling, or algorithmic topology searches, significantly minimizing maximum neural capacity relative to potential.
- **Hardware Restrictions:** Evaluation scaled limited subsets manually resulting in minimal graphical processing pressure.

VIII. Conclusion and Future Explorations

Conclusively, this study verified that standard quantitative forecasting implementations should definitively be dynamically evaluated based fundamentally on existing parameters spanning length capabilities. Small datasets mandate adoption characterizing traditional simplistic methods maintaining low training variance penalties. Ultimately however, maximizing deployment architectures into modern sequence models like LSTMs remains structurally mandatory exclusively once extensive archival databases exceeding multiple thousands of occurrences allow computational capability scaling.

Subsequent investigations generated outward from this operational proof should functionally expand testing utilizing GRU variants juxtaposed explicitly against multi-dimensional transformer architectures enforcing dynamic contextual self-attention over immensely expanded big data metrics targeting global macroeconomic shifts rather than isolated static sequences.

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